Mishil Trivedi

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EDUCATION

M.S. in Business Analytics (STEM) **January’26**
Boston University, Questrom School of Business, Boston, MA GPA: 3.52
B.A (Hons) Economics **July’24**
Delhi University, Delhi, India

RELEVANT WORK EXPERIENCE

Data Engineer (Volunteer) — *NAAAP* *September’25 – Present*
- Engineered automated ETL pipelines using Python and SQL to ingest, clean, and unify multi-source engagement data, processing 10K+ records/month into analytics-ready tables.
- Designed normalized data models and transformation logic in dbt to standardize event, user, and interaction data, improving downstream query performance and data reliability by 35%.
- Collaborated with analytics and stakeholder teams to translate reporting requirements into scalable data structures, reducing analysis turnaround time by 25%.

Principal Data Engineer — *SALT* *May’26 – Present*
- Built a Boston restaurant API on Cloud Run with a 5-layer extraction pipeline (JSON-LD, Playwright, Gemini, search fallback) to work in under 2 seconds
- Created AI cost structures to maximize Cloud Server efficiency with low latency while 8+ hours per week.
- Shipped the B2B pilot end-to-end and wrote API docs, built consumer test scripts, triaged six bugs across auth, latency, and shape, with the aim of ensuring the consumer integrates the API successfully.

PROJECT EXPERIENCE

Multi Agent AI Chart Understanding & Transformation System *September’25 – December’25*
- Built a data extraction and transformation pipeline to convert visual chart inputs into structured tabular representations, enabling consistent downstream analytics across heterogeneous chart formats.
- Engineered an automated code generation and validation workflow using Gemini 2.5 Pro to regenerate analytics-ready visualizations, achieving over 90% successful render accuracy.
- Implemented automated error recovery, schema checks, and metric consistency validation, reducing end-to-end transformation latency by 3 times.

Job Market Analytics Platform *September’25 – December’25*
- Architected six end-to-end data pipelines (ingestion, enrichment, embeddings, entity resolution, clustering, matching) using Python and SQL, improving refresh cadence from weekly to daily with 100% pipeline reliability.
- Developed LLM- and spaCy-based enrichment services to extract skills, seniority, and role context from unstructured job data, increasing structured feature coverage by 40%.
- Implemented fuzzy matching, rule-based normalization, and validation checks to resolve entity inconsistencies, improving company-level matching precision by 35% and eliminating 15+ hours/week of manual cleanup.

Voice-to-SQL Agent *February’26 – April’26*
- Engineered a local-first, autonomous voice-to-SQL AI agent using Whisper and quantized Llama 3/Qwen models to query massive enterprise datasets with zero latency and 100% data privacy.
- Implemented a self-healing execution loop that autonomously catches database schema errors and prompts the LLM to rewrite broken SQL, eliminating manual debugging.
- Implemented drift detection and performance monitoring using statistical diagnostics and SQL-based alerting, improving model reliability under distribution shift.

SKILLS

- **Languages & Querying:** Python, SQL.
- **Data Engineering:** ETL/ELT, Data Pipelines, Data Modeling, Schema Design, Data Validation, Entity Resolution, API.
- **Processing & NLP:** Pandas, spaCy, LLM Embeddings, Semantic Enrichment.
- **Infrastructure Concepts:** Pipeline Orchestration, Fault Tolerance, Data Quality Auditing.
- **Tools:** Git, Jupyter, SQL Databases, LLM APIs.
- **Analytics Foundations:** Feature Engineering, Clustering, Similarity Search, Normalization.